

Processors

How would you like to customise your processor?

Processors! The very word brings to mind numerical crunching monsters that eat a few thousand floating points for breakfast. This is one dominion of computing that enjoys big increments in performance every year or so. Ever-shrinking fabrication processes bring the all-significant advantages of reducing costs (hence prices) and power savings.

The two giants, Intel and AMD, are locked in a struggle for top market dog. We've witnessed the fallout courtesy AMD's Athlon 64 a few years ago. More recently Intel's Core 2 Duo (code-named Merom for laptops) fuelled retaliation; it came as a shock, even to industry experts. The Gigahertz battle is a thing of the past, and parallelism (read multiple cores) is the new order of things.

Intel's Core 2 Duo desktop and notebook range will soon be replaced by a derivative of the same architecture code-named Penryn.

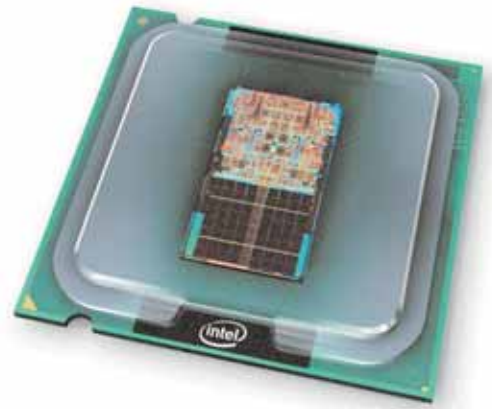
Penryn gets a die shrink from 65nm to 45nm. Code-named Wolfdale, the dual-core Penryn features 6 MB of L2 cache (opposed to the Core 2 Duo's 4 MB). The quad-core Penryn, code-named Yorkfield, adds 4 MB over the Core 2 Quad's 8 MB L2 cache.

Besides minor architectural changes like reduction in processor instruction latency, improved IPC (Instructions Per Clock), and tweaks to the cache and memory management subsystems, Penryn's improved thermal efficiency allows it to enjoy higher clocks than Core 2. Improvements have also been made in power management, but this is restricted to the notebook space, where thermal envelopes are more restrictive.

Intel's benchmarks indicate nearly 25% performance increments over Core 2. While such a large boost is exciting, there's even more... Nehalem, a radical new architecture as Intel pitches it, takes the term "multithreading" to a whole new level. With native support for eight cores on a single die (the Core 2 micro-architecture could handle four cores tops), Nehalem has the industry salivating. Although details of power management are sketchy, Intel quotes "dynamic management," which could mean independent power management for each core.

Nehalem goes the Athlon 64 way with an integrated memory controller (IMC). For those not in the know, the memory controller in Intel systems is situated on the Northbridge, while AMD had integrated it on the CPU die with their first 64-bit Desktop processors. This kind of architecture cuts down on latent clock cycles, thereby improving memory-related performance. Another leaf out of AMD's book: Nehalem uses a point-to-point serial interconnect instead of an FSB (AMD calls this HyperTransport technology).

Something new is a hybrid of CPU and GPU (GPU on the CPU die itself)! This is Intel's version of AMD's "Fusion" technology. Fusion is an inter-



esting concept that modularises processors. Both AMD and Intel have showcased Fusion as a computing evolution. It also brings flexibility to the manufacturing process.

Imagine—manufacturers are manufacturing a processing solution based on three CPU cores and one GPU core. Suppose demand for gaming suddenly increases. Manufacturers have the option to simply plonk more 3D processing cores onto a single wafer, say three graphics cores and one general processing core, thus saving on material and manufacturing costs tremendously. Nehalem will require a new platform due to the large number of non-backward-compatible changes.

AMD's been on a bit of a rollercoaster since tweaking Intel's tail back in early 2006. Disaster control plans revolve around a successor architecture to replace K8 (Hammer) powering their Athlon 64 and AM2 processor families. Barcelona is the codename for their latest architecture (65nm), which will be tasked with swiping back the performance crown lost a year back. It is AMD's first (native) quad-core. It has a big advantage as opposed to Kentsfield, which is basically a dual, dual-core solution (therefore not native). Barcelona features an optional L3 cache, and wide execution—128-bit to be exact, so 128-bit SSE (Steaming SIMD Extension) operations are possible with a single micro-operation. SSE allows processing of

multiple data streams by a single command. This sort of execution works very well for audio and video encoding as well as image editing operations. AMD calls this SSE 128. The K8 had a 64-bit execution engine, which meant any 128-bit operations had to be decoded into two 64-bit micro-operations, resulting in wasted clock cycles.

